

the volume of data provided for discovery, but also by the quality of the data.

Pattern discovery: An overview

In today's dynamic market, conventional business intelligence often falls short in enabling the discovery of actionable insights from the huge volumes of data available. Let's consider the case of the retail industry where the consumer trends change very rapidly. A single social media post from an influential trendsetter may cause a considerable amount of disruption in the market. Hence, the retailers need to have real-time insights to ensure that all their plans and forecasts are relevant. In this situation, techniques like smart pattern discovery enable automation and acceleration of the process of finding trends and predicting the behaviour of the consumer.

While this technology is being quickly adopted in the retail domain, pattern mining and pattern discovery may be very useful in many other domains. Following this trend, many leading commercial vendors are enhancing their tools with the capability of discovering inconspicuous data patterns hidden in the enterprise data store. As many of these tools and platforms are quite expensive, it makes sense to take a look at the tool-sets available in the open source domain to pick up the appropriate one.

Even with the wide range of algorithms and tools available, effective use of pattern discovery calls for a suitable implementation methodology.

Implementation methodology

Discovering patterns related to sequences of data or events can lead to very important insights, as these patterns may have occurred before an event of significance. If such patterns can be identified, it may be possible to proactively track them and take appropriate action in a timely manner. Due to the increase in data volumes, it is no longer possible for subject matter experts to scan the data and identify significant patterns. Leveraging suitable machine learning algorithms can be instrumental in reducing the time and effort humans would need to put into this task.

While there is a huge need to identify the hidden patterns in the data set across various business domains and there are a handful of tools and techniques to address this, it is always required to build a systematic implementation methodology for effective execution and to obtain the best



Figure 1: How the algorithm works

results. The methodology has to be a combination of a structured approach as well as the right choice of tools and algorithms.

The frequent pattern mining (FPM) algorithm

Finding a pattern from a huge and dynamic enterprise data store typically takes up a considerable amount of time and effort. Alternatively, this task has to be performed using suitable machine learning algorithms like FPM and other related algorithms.

As depicted in Figure 1, pattern mining algorithms like FPM take token sequences as input, where a token indicates elements of significance. For example, a token can be an error code, a transaction type or an item from a shopping cart. On being provided the token collection, these algorithms try to identify the most frequent tokens and token sequences with the associated probability, based on the occurrence of the tokens in the complete set (transaction database).

Tool-set in open source

Till recent times, machine learning was confined more to the academic world. This may be the reason for the vast profusion of open source tools and libraries around machine learning and related technologies.

Some of the popular tools are R, Python, Apache Mahout, H2O and Rapid Miner. While each of these tools is created to cater to a specific set of requirements, R provides good support for FPM implementation.

Framework

While FPM algorithms are quite effective for pattern mining or pattern discovery, a systematic implementation approach ensures effective execution and the best results.

An enterprise that wishes to use FPM techniques would have to implement a framework as depicted in Figure 2. The FPM solution will need to have a data store that has the ability to store data from a variety of sources. The framework also needs an analytics engine that can operate on the data to identify frequently occurring patterns to

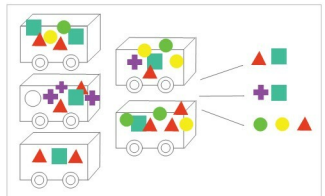


Figure 2: Framework architecture